

SAFMR

Smart Autonomous Floor Mopping Robot

It maps the room, plans the cleanest path across it, and sweeps, mops and vacuums — without anyone pushing it there.

WHY WE BUILT THIS

05:45 AM



Same floor. Same mop. Every day.

Across offices, hospitals and hotels, someone starts their shift by pushing a mop over the same square metres — again.

40%

MANAGE 5-15K SQ FT

7

DAYS A WEEK, ON REPEAT

0

DAYS IT CLEANS ITSELF

"It's not that the job is hard. It's that it never ends."

THE QUESTION



What if the floor cleaned itself?

45% of facility managers we surveyed call labour cost and staffing their single biggest cleaning headache.

35%

say the results are just inconsistent — it depends who's mopping that day

15%

say cleaning simply takes too long to get through

5%

flag staff safety on wet or busy floors

Every answer pointed the same way: less mop, more machine.

MEET SAFMR

It scans, plans, and gets on with it.



The floor cleaning problem

From a survey of 20 facility managers across offices, hospitals and hotels

45%

say labour cost and staffing is their single biggest cleaning challenge

65%

are already looking to buy or rent a commercial cleaning robot

95%

would consider buying at around LKR 200,000

80%

rate cutting cleaning labour hours as very important (6-7 of 7)

The gap: capable autonomous scrubbers exist, but only as imported units priced far out of reach for most local facilities.

What SAFMR does differently

One machine, one pass, no supervision required

01 Sees the whole room, not just the next step.

LiDAR mapping with ROS2 Nav2 and SLAM builds a real floor plan and plans an efficient route across it.

02 Sweeps, mops and vacuums — in one pass.

No separate machines or extra passes: brushes, counter-rotating mop discs and vacuum pickup all run together.

03 Leaves almost nothing untouched.

A serpentine coverage path reached ~94% of the reachable floor area in testing — no skipped corners.

04 Knows where not to go.

Cliff and obstacle sensing stops it at stairs and edges, and steers it around people and furniture.

05 Runs from your pocket.

Start a clean, check progress, or call it back to dock, all from a simple mobile app.

45 cm CHASSIS · DUAL MOPPING DISCS + SWEEP BRUSHES · ~94% COVERAGE · ~60 MIN RUNTIME / CHARGE · RASPBERRY PI 5 + STM32F407 · ANDROID APP CONTROL · ~LKR 200,000 TARGET PRICE



SAFMR — Team MindFlayers

Built for institutions that are done paying, every single day, for someone to clean the same square metre of floor.